

Virtual Agent–Driven SLA Breach Awareness & Justification System

Phase 1

Brainstorming & Ideation

Problem Statement:

Many organizations struggle with Service Level Agreement (SLA) violations because assignees are often unaware that incidents are approaching breach deadlines. Manual follow-ups, delayed communication, and lack of accountability reduce service quality and increase operational risks. This project develops a **Virtual Agent–Driven SLA Breach Awareness & Justification System** in ServiceNow that proactively alerts assignees before SLA breaches, captures justification through Virtual Agent conversations, and maintains audit-ready records for governance.

Empathy Map:

Target User: IT Support Agent responsible for handling incidents. Users need timely reminders, an easy way to submit breach reasons, conversational support, and better visibility into SLA deadlines.

Phase 2

Requirement Analysis

Technology Stack:

Platform: ServiceNow

Automation: Flow Designer

Virtual Agent: ServiceNow Virtual Agent

Notifications: Email & Flow Designer

Reporting: Dashboards & Reports

Security: ACLs, Roles & UI Policies

Customer Journey:

Incident Assignment → SLA Monitoring → 80% SLA → Notification → Virtual Agent → Justification → Incident Update → Manager Review

DFD:

The system monitors incident SLAs, triggers notifications near breach, collects justification via Virtual Agent, updates the incident, and stores information for reporting and audits.

Phase 3

Project Design

Problem–Solution Fit:

- Automatic SLA monitoring
- Proactive notifications
- Virtual Agent justification capture
- Automatic work notes
- Reporting dashboards

Architecture:

Incident Table → SLA Engine → Flow Designer → Virtual Agent → Incident Update → Reports & Dashboards

Phase 4

Project Planning

Milestones:

1. Requirement Analysis
2. Backend Configuration
3. Flow Designer Development
4. UI Policy Creation
5. Virtual Agent Design
6. Reporting & Dashboards
7. Testing
8. Deployment

Development followed sprint-based planning.

Phase 5

Project Development

Backend: SLA configuration, Flow Designer, Notifications, UI Policies, Business Rules.

Virtual Agent: Greeting topic, SLA awareness, justification capture, incident update, confirmation. Implemented proactive alerts, automated work notes, justification tracking, and audit-ready records.

Phase 6

Project Testing

Testing covered SLA calculations, Flow Designer execution, Virtual Agent conversations, incident updates, notification delivery, dashboard validation, and security verification. All functional test cases passed successfully.

Phase 7

Project Documentation

Documentation includes business requirements, functional requirements, ServiceNow configuration guide, Flow Designer setup, Virtual Agent setup, UI policy configuration, testing report, user manual, and administrator guide.

Phase 8

Project Demonstration

Demonstration includes incident creation, SLA monitoring, automatic notifications, Virtual Agent interaction, justification submission, incident updates, and dashboards.

Future Scope:

- AI-based SLA Risk Prediction
- Microsoft Teams & Slack Integration
- Predictive Analytics Dashboard
- GenAI-based Justification Suggestions
- Mobile Push Notifications
- Multi-language Virtual Agent